Representation Learning of Time-Series Complexity for Detecting Life and Technology

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Abstract

We trained a small convolutional neural network as a time-series encoder and used contrastive learning so that windows from the same *type* of measurement tend to land nearby in a 64-dimensional embedding space. The training mix spans abiotic environmental and geophysical series, biotic physiological recordings, and synthetic or instrument-heavy technological traces (see Table 1). After training, PCA and t-SNE projections show that the three coarse classes partly separate and that individual signal families form recognizable neighborhoods. This is a course-scale proof of concept: we are not proposing a field-ready detector, but the learned geometry is consistent with the idea that machine learning can reveal organization in time-series data without fixing a single hand-crafted complexity score in advance.

1 Introduction

A defining challenge in astrobiology is our lack of priors and constraints on what all relevant signatures of life or technology might look like. Classical technosignature searches often focus on specific signal morphologies (e.g. optical laser pulses [1] and narrow-band radio transmissions [2]) that are easier to motivate physically and distinguish from noisy baselines. More agnostic biosignature work instead asks whether living systems generate observables with distinctive forms of complexity or variability. While both directions are compelling, each inserts different assumptions into the analysis pipeline.

This project operates between these two traditions. Rather than starting from a hard-coded signal template or a single hand-crafted scalar notion of complexity, we ask whether a model can learn a useful representation of time-series structure directly from data. Our intuition is physically motivated: if abiotic, biotic, and technological systems produce measurably different temporal organizations, then one should be able to construct an embedding space, learned from signals, in which those differences emerge as geometry. This framing is intended to leverage the expressivity of modern deep learning architectures through an end-to-end unsupervised approach.

1.1 Machine learning for technosignature detection

Machine learning is already integrated into modern SETI pipelines, as the observational regime is both large-scale and extremely sparse: radio telescopes produce vast numbers of spectrograms, while true candidates are expected to be extremely rare, a "needle in the haystack" problem. At this scale, modest but reliable automations have an impact, so many existing methods are designed around a specific target class, usually structured narrowband radio signals that drift in frequency from the acceleration between the transmitter and receiver, due to the orbits and rotations of both.

In 2020, the Breakthrough Listen Initiative treated SETI observations from the Green Bank Telescope (GBT) as time-frequency images and trained convolutional neural networks (CNNs) to localize

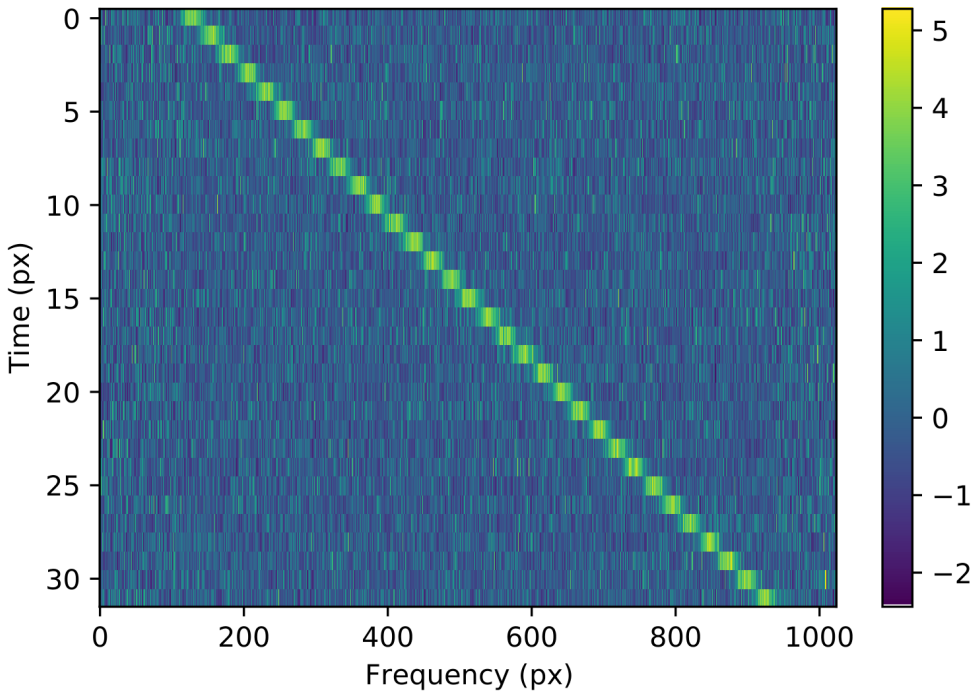


Figure 1: Synthetic spectrogram frame from [3], showing two narrowband signals: a bright radio-frequency interference (RFI) signal at 25 dB with zero drift, and a dimmer drifting signal at 15 dB. This shows the structured spectrogram morphology targeted by many SETI pipelines.

signals in noisy synthetic spectrograms. They showed that residual CNNs can successfully identify structured signals even in crowded frames (example in Fig. 1), and outperform simpler baselines, particularly when signals are dim or overlap with brighter interference [3]. However, the success of this proof of concept relied on a strong prior on the structure of the signal. While scientifically reasonable for many SETI searches, this is a template-driven problem, as their model learned to recognize a specific family of morphologies, not a general notion of organized behavior.

In 2023, the same initiative scaled up this approach, applying deep learning to GBT observations of 820 nearby stars, totaling more than 480 hours of telescope time. Their pipeline used a β-variational autoencoder to learn the structure in spectrogram data and separate candidates from radio-frequency interference (RFI), and out of roughly eight million detections, the search returned eight promising signals to further observe and monitor [4]. Most recently, this data-driven approach was extended into a full anomaly-detection pipeline, analyzing 10^{11} spectrograms from Parkes Observatory and GBT, ranking candidates according to criteria like frequency rarity and signal persistence in time. They then used dimensionality reduction, including UMAP from 15 dimensions to 2, to visualize and triage the resulting candidate space before human inspection, but after inspecting the 20,000 candidates, none survived basic human scrutiny. We note the use of dimensionality reduction here, compressing features into a space where one can begin to assign and derive meaning in a lower-dimensional and more interpretable space [5].

We notice that modern ML methods can detect structure, compress large search spaces, and make SETI pipelines computationally tractable, but they expose a shared limitation. Pipelines either (i) assume the signal resembles an engineered template or (ii) define "interestingness" through deviation from a background distribution built from existing observations, both inheriting assumptions about what defines a signal as noteworthy. In astrobiology, important signals may be out of distribution in ways that are scientifically meaningful but not easily expressed by a single template or anomaly score. This project is less interested in whether a model can find one canonical technosignature class, and more interested in whether a model can learn an underlying representation of time-series organization that distinguishes broader classes.

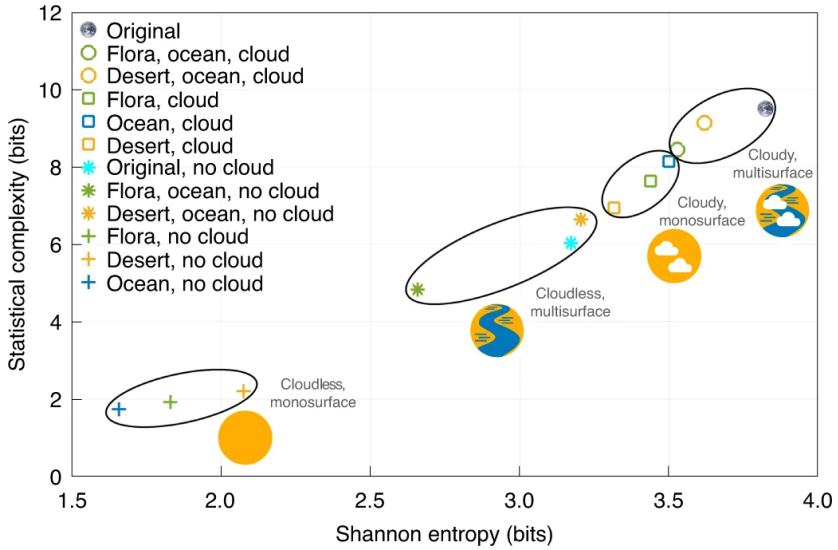


Figure 2: Schematic complexity–entropy diagram from [7], illustrating the intuition of statistical complexity as a function of Shannon entropy rate and introducing a 2D interpretation of a complexity space.

1.2 Complexity as an agnostic biosignature

Life itself tends to generate observables with characteristic complexity, since living systems are neither perfectly ordered nor perfectly random: they show structured variability, and in some cases, long-range correlations in time [6]. The organization of a signal may itself carry evidence about the nature of its source. Bartlett et al. apply this idea to create a computational framework that models time-series data with reconstructed ϵ-machines, minimal predictive models that capture the causal structure required to forecast future observations from the past [7]. In this setting, statistical complexity measures the information required to represent that predictive structure. Intuitively, purely random processes might be difficult to predict but require little internal model structure, while perfectly periodic processes are highly ordered and also simple. The most interesting systems exist between these limits: neither trivially random nor perfectly predictable.

This framework is paired with the argument that planets hosting biospheres may therefore occupy high-complexity regions of observational signal space, making complexity itself a plausible biosignature candidate, a framing useful to this project, as it shifts attention toward organization itself. Recall that in the time-series challenge presented in the final assignment of Ge/Ay 159, the goal was not only to identify intentional technological transmissions but also to reason about intuitively, naturally generated time series from biotic sources. In that setting, hand-crafted metrics like entropy and compression ratios helped students interpret structure, and helped operationalize the intuition that signals produced by a living or organized system might occupy different regions of a "complexity space." But explicit complexity metrics impose limitations. First, each metric encodes a choice about what kind of structure matters: Shannon entropy emphasizes unpredictability, compression ratios reward redundancy, and autocorrelation measures certain time scales. Second, many such metrics reduce a complicated time series into a single scalar, which may be informative, but erases any multi-scale nuance. It is often not a sufficient statistic, let alone a minimally sufficient statistic. Third, and most important for astrobiology, these metrics introduce bias. By choosing a notion of complexity, we capture structures familiar from Earth systems while missing or downweighting out-of-distribution patterns. Defining complexity too rigidly undermines our goal of maximizing agnosticity.

For these reasons, we seek a method that remains sensitive to temporal organization without forcing that organization to be measured by one statistic. That is, complexity matters, but explicit complexity metrics are incomplete and potentially biased, and we ask: can we instead learn a representation of complexity from raw time-series data?

1.3 Contrastive Representation Learning

Representation learning is a subfield of machine learning that is well positioned to address the question above. It aims to map raw data into a higher-dimensional latent space in which the geometry captures meaningful features. For time-series data, this means learning an encoder $x_{1:T} \mapsto z \in \mathbb{R}^d$ such that distances or neighborhoods in the latent space reflect similarities in the original signals. This is attractive in our setting because it removes the need to define features manually in advance. Rather than computing one entropy score, one compression score, or one periodicity score, and then assembling them either into a vector or into a scalar through some uncertainty-weighted linear combination, we allow the model to infer which parts of the waveform are useful for organizing the data.

Franceschi et al. provides an important precedent for this approach in the time-series literature, showing that neural encoders can learn general-purpose time-series embeddings through a contrastive objective built from positive and negative samples [8]. Their model is trained with a triplet-style loss (shown in Fig. 2): a reference signal and a related "positive" signal are encouraged to map nearby in latent space, while unrelated "negative" segments are pushed farther away. Note that this does not require manually labeling each input by its semantic source class. Instead, the supervision comes from the structure of the sampling procedure itself: different windows drawn from the same underlying signal are treated as positives, while windows drawn from different signals are treated as negatives. For example, two snippets of the same heartbeat time series would be encouraged to lie close together in the embedding space, while two snippets of an earthquake trace would also cluster together, even though the heartbeat and earthquake clusters would lie far apart from one another. In this way, the model learns a relational geometry of similarity directly from the data, rather than from explicit source labels. Similar signals cluster, dissimilar signals separate, and the resulting representation supports downstream analysis even when labeled data are limited or unavailable.

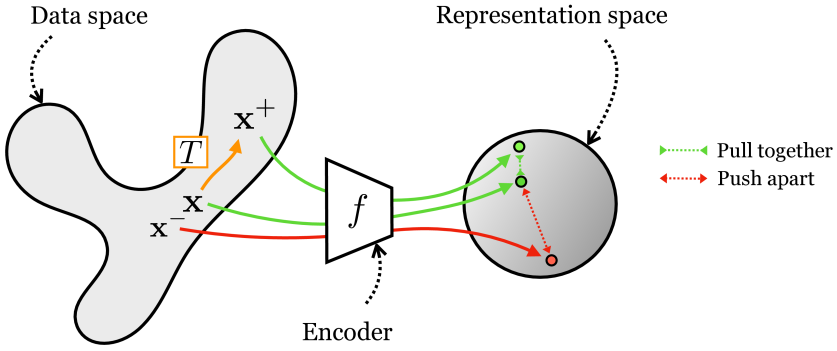


Figure 3: Contrastive representation learning for time-series data, as shown in [9]. A neural encoder learns a mapping from raw signal segments $x_{1:T}$ to latent representations $z \in \mathbb{R}^d$ such that positive pairs (x, x^+) are pulled together in representation space while negative pairs (x, x^-) are pushed apart.

A contrastive objective does not require us to say, in advance, why two segments of a time-series are similar. It only requires a principled way to define which segments should be semantically close and which should be treated as distinct, a much better fit for our problem. Windows drawn from the same source time series can serve as positives, and windows form different sources serve as negatives, so the encoders is encouraged to discover the features that make those relationships consistent.

The resulting pipeline is as follows: we begin with time-series signals from abiotic, biotic, and technological sources. We divide them into segments, pass those segments through a neural encoder during training, and eventually obtain latent vectors for each one. After training, we analyze the learned space using dimensionality reduction methods such as PCA and t-SNE to probe whether broad process classes occupy different regions. This is distinguished from earlier technosignature work, as the goal is not to immediately declare detections or rank candidates. We are not building a production search pipeline, nor are we claiming that latent clustering by itself establishes the presence of life or technology. Instead, the learned representation itself acts as a tool not only for filtering, but also for interpretability. If the embedding systematically reflects different forms of organization, it provides a less assumption-heavy way to identify ambiguous signals and reanalyze clustered signals

to refine our understanding of useful summary statistics. From modern ML technosignature searches, we take the lesson that neural networks can scale to large, noisy, high-dimensional data. From agnostic biosignature and complexity work, we take the lesson that organized systems may reveal themselves through the structure of their variability. Through contrastive representation learning, we allow complexity to emerge as a learned geometry rather than a predefined statistic.

2 Method

Scientific goal of the pipeline

The method is designed around an astrobiological question posed in Ge/Ay 159: if we hope someday to flag *organized* or *complexity-structured* behavior in alien data, how can we compare signals without deciding in advance that life or technology must look like a single template? Classical technosignatures often fix the hypothesis (narrowband radio, laser pulses). Agnostic biosignature work instead emphasizes statistics of variability. Our pipeline sits between those poles: we still use Earth data as a convenience laboratory, but we ask a neural encoder to *learn* what "similar temporal organization" means across domains, then we inspect the geometry with standard visualization tools. The point is exploratory: we want to see whether abiotic, biotic, and technological traces naturally occupy different neighborhoods of a learned space, before anyone attaches detection thresholds or mission requirements.

Assembling new data for this study

We did not use a single off-the-shelf benchmark. Instead we assembled a multi-source corpus listed in Table 1, combining public environmental and geophysical feeds, curated clinical-style physiological archives, and purpose-built technological stand-ins (synthetic RF traces, Morse-like waveforms, machine vibration envelopes, and public "anomaly benchmark" telemetry streams).

Table 1: Time-series sources used in this study, grouped by the coarse label supplied to the class-colored plots.

Class	Signals
Abiotic	Solar flares; river flow; global temperature; earthquake seismograms.
Biotic	Heart rate; respiration; EEG.
Technological	Morse code; RF communication signals; machine audio envelopes; AWS telemetry.

Windowing and training protocol

Each continuous recording is cut into overlapping windows of 256 samples with stride 128, z-scored inside each window so amplitude units do not dominate comparisons, and assigned to train, validation, or test with the rule that all windows from the same underlying record stay in the same split. That choice matters astrobiologically: it mimics the idea that we should not "train" on one portion of a planetary dataset and naively *test* on adjacent segments of the same instrument run and call it independent evidence.

The encoder is a **one-dimensional convolutional network**: stacked convolutions and pooling sweep along the time axis, then a linear map produces a 64-dimensional embedding normalized to unit length so that inner products behave like cosine similarities. Training is **contrastive**: for each window we synthesize two augmented views (Gaussian noise, random rescaling, small circular time shift) and push their embeddings together when—and only when—they share the same *fine* signal identity (separate RF modulation families count as distinct types). The loss averages over minibatches; we optimize with Adam (10^{-3} learning rate) for 20 epochs. The descending loss curve is our primary quantitative sanity check that the network is performing a nontrivial calculation rather than memorizing filenames. After training we freeze the weights, embed every retained window, and apply **PCA** (two components, global geometry) and **t-SNE** (local clustering, perplexity 30, PCA-initialized) to obtain the figures in Section 3.

3 Results

Figure 4 shows PCA and t-SNE side by side with points colored by abiotic, biotic, or technological class; Figure 5 repeats the layout colored by individual signal family. Training ran for 20 epochs with decreasing loss and produced thousands of embedded windows from Table 1.

The class-colored maps show that PCA captures a curved global arrangement: abiotic, biotic, and technological populations occupy partly distinct zones (technology often in dense clusters on one side of the cloud, abiotic along the opposite flank, biotic in a middle band), with overlap where biology and environment share stochastic structure. t-SNE emphasizes tighter local neighborhoods and makes technological windows look the most internally coherent, abiotic the most heterogeneous, and biotic more fragmented in between—consistent with mixing many unrelated physical processes versus repeating engineered modulation. The signal-family panels add that RF, machine audio, and telemetry form tight islands, and EEG often lies closer to technological regions than heart rate or respiration: although we labeled EEG as biotic, its oscillatory, band-limited, sensor-shaped traces resemble technology-like regularity, which the embedding surfaces rather than hiding it beside other physiology.

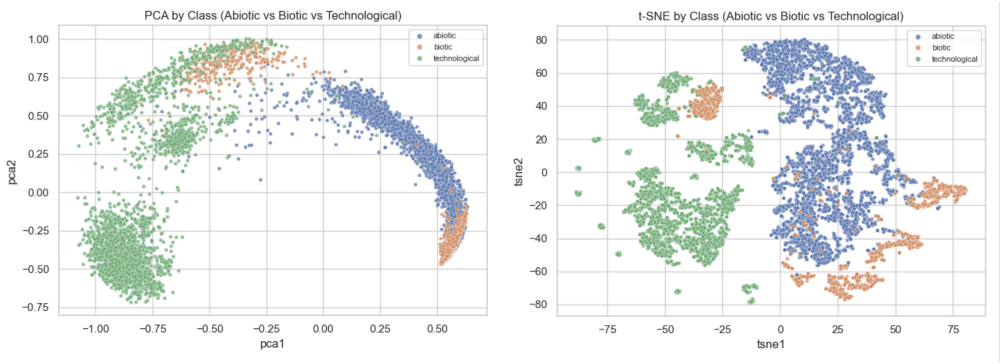


Figure 4: Embedding vectors projected to two dimensions. **Left:** PCA, colored by coarse class (abiotic, biotic, technological). **Right:** t-SNE with the same legend. The curved PCA cloud indicates that the 64-D representation is not perfectly linear; t-SNE highlights tighter technological density versus more spread abiotic neighborhoods.

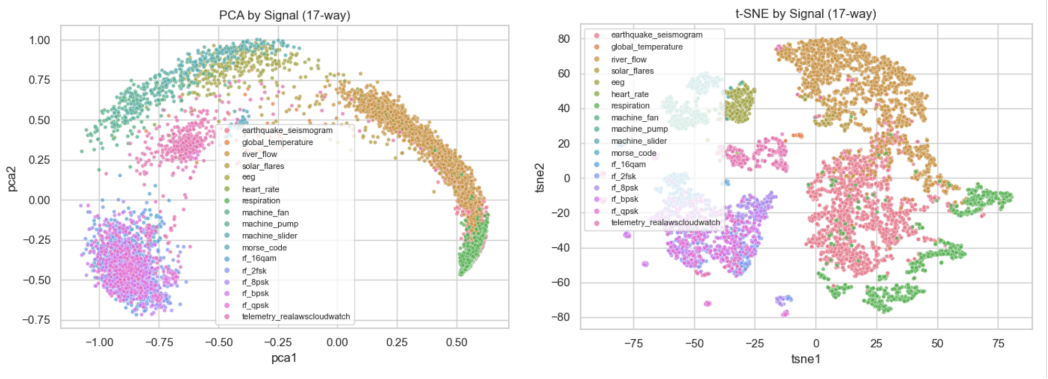


Figure 5: **Left:** PCA colored by signal family (17 types). **Right:** t-SNE with the same legend. Tight islands for RF modulations, machine audio, and telemetry indicate that the encoder picks up modulation, rhythm, and noise texture. EEG often appears closer to technological clusters than other biotic traces: although we labeled EEG as biotic, its oscillatory, band-limited, sensor-mediated waveform resembles engineered or protocol-regular signals, and the embedding reflects that tension rather than forcing it beside heart rate or respiration.

4 Division of labor

Both authors contributed equally. We jointly framed the project (representation learning as a complement to hand-tuned complexity metrics), implemented the machine-learning experiment end to end, produced the embedding figures, and wrote this paper together.

5 Conclusion

Where much technosignature machine learning still attaches to a pre-imposed pattern (e.g., localized structures in spectrograms), our experiment asks whether a single encoder can carve out a shared notion of temporal organization across *unrelated* Earth analogs and still respect the astrobiologically motivated trichotomy abiotic/biotic/technological. The geometry we obtain supports three interpretive claims: (i) the model finds coarse class structure without being told only those three labels at train time; (ii) technological stand-ins are the most internally coherent, plausibly because modulation and protocol impose repeated motifs rare in planetary noise; (iii) biotic traces occupy an intermediate regime, mixing stochastic elements of natural environments with quieter regularity than narrowband radio, and (iv) the embedding surface label tensions—for example EEG clustering nearer technological neighborhoods reminds us that "biotic" in a mission context may need to factor in how the measurement is made, not only the organism.

To conclude, for this project we integrated contrastive representation learning into the astrobiological discussion of agnostic signatures, we performed a concrete multi-source data and training exercise, and we extracted qualitative lessons (class geometry, technology coherence, EEG versus other biotic traces) that inform how future pipelines might be designed and critiqued.

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